Rafael Henrique Mori Belmonte

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Education

Minerva University — **San Francisco, CA**
Bachelor of Science, Major in Computational Sciences, Minor in Neuroscience — *Expected May 2028*

- Selected for global undergraduate program (1.2% acceptance rate) with studies across four continents.
- Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Statistics, Probability, AI, ML, Neuroscience

ETEC Presidente Vargas — **São Paulo, Brazil**
High School and Technical Diploma in Electronic Engineering, GPA 4.0/4.0 — *2023*

- Relevant Coursework: C/C++, Embedded Systems, Microprocessors, Electronics, Automation, IoT, Circuit Design.

Experience

Minerva University — **San Francisco, CA**
Math Research Assistant & Software Engineer Intern — *Aug 2024 – Present*

- Developed 2D Arbitrary Lagrangian–Eulerian (ALE) and Finite Element Method (FEM) simulations to model symmetric impacts and analyze dynamic responses in elastic materials.
- Built a full-stack assessment platform with React, Next.js, TypeScript, and Firebase, implementing secure authentication, dynamic questionnaires, and psychometric matching algorithms for ~150 incoming students annually.

NTT East — **Tokyo, Japan**
Research & Development Engineer Intern — *Jan 2026 – Apr 2026*

- Designed BOUS.AI, an AI-enabled disaster response application integrating GIS data, shelter management, and multilingual alerts to support emergency coordination across 17 Japanese prefectures.
- Led product development from stakeholder requirements to prototype, incorporating NTT East feedback and presenting a business model and implementation proposal to company leadership.

NERU-box — **São Paulo, Brazil**
Co-Founder — *May 2025 – Present*

- Lead a 4-person team developing an assistive robot and React Native app integrating task management, parental controls, and research-informed reward systems for neurodivergent users.
- Architect hardware/software systems featuring privacy-focused AI assistance, computer vision-based task verification, behavioral analytics, and personalized recommendations.

Projects

Tiqueteiro — **São Paulo, Brazil**
Project Manager — *Feb – Aug 2023*

- Built a full-stack ticketing platform using MERN stack and Typescript, integrating email verification and QR-code ticketing for a 2,000+ attendee event while reducing expenses by 40% (R$4,000).

Honors

- Selected for the 2026 Uber Career Prep Fellowship (<4% acceptance rate), receiving software engineering mentorship, technical interview preparation, and industry training.
- Awarded 2 Silver Medals in the Brazilian Mathematics Olympiad (OBMEP), ranking among the top 0.01% of 18 million participants nationwide.
- Earned 4 medals (2 Gold, 1 Silver, 1 Bronze) in the Brazilian Astronomy and Astronautics Olympiad (OBA).

Skills

- **Programming:** Python, C/C++, JavaScript, TypeScript, SQL
- **Frameworks & Tools:** React, React Native, Next.js, Node.js, Express.js, Expo, Firebase, MongoDB
- **Hardware & Research:** ESP32, Arduino, PCB Design, Embedded Systems, IoT, FEM/ALE Modeling
- **Languages:** Portuguese (Native), English (Fluent), Spanish (Advanced), Japanese (Elementary)